Project 2 - Climate dynamics

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Part 1

1)

To find an analytic solution to the initial value problem using the eigenvalue-eigenvector method for the system

$$x' = 4x - y,$$
$$y' = 2x + y$$

we need to find the eigenvalues and eigenvectors for the matrix

$$\begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}$$

The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$ and the corresponding eigenvectors are $e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $e_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

This means that the solution is

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t}$$

We can also find a numerical solution to the initial value problem using a numerical ODE solver

```
function dydt = ode1(t, y)
    dydt = [4*y(1) - y(2); 2*y(1) + y(2)];
end

[t,y] = ode45(@ode1,[0 1],[a; b]);
```

where $[a; b]$ is the starting point.

We can compare the two solutions on a plot

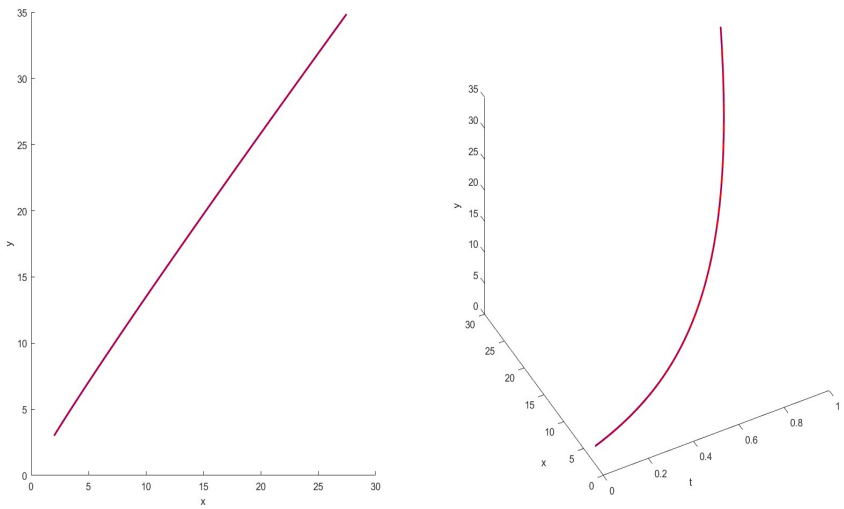


Figure 1: Solutions of the first linear system using as starting point (2,3). In blue the analytic solution, in red the numerical solution. They are completely overlapping. The color scheme will be the same for all the other plots of this exercise

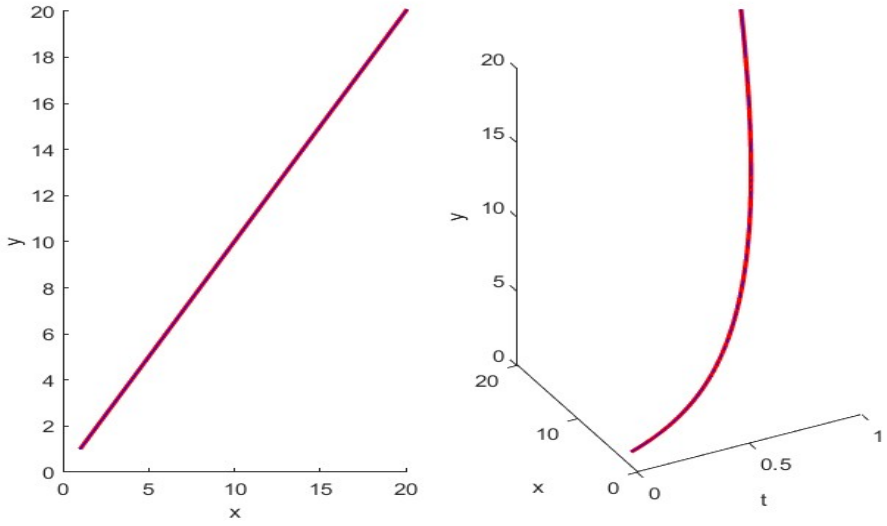


Figure 2: starting point (1,1)

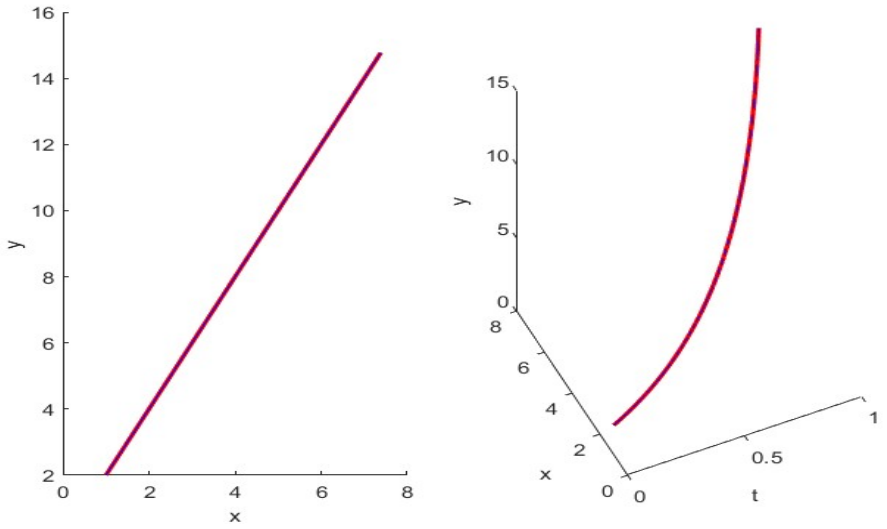


Figure 3: starting point (1,2)

As we can see, the solutions are exactly the same.

The phase portrait centered on the fixed point at the origin is in fig.4

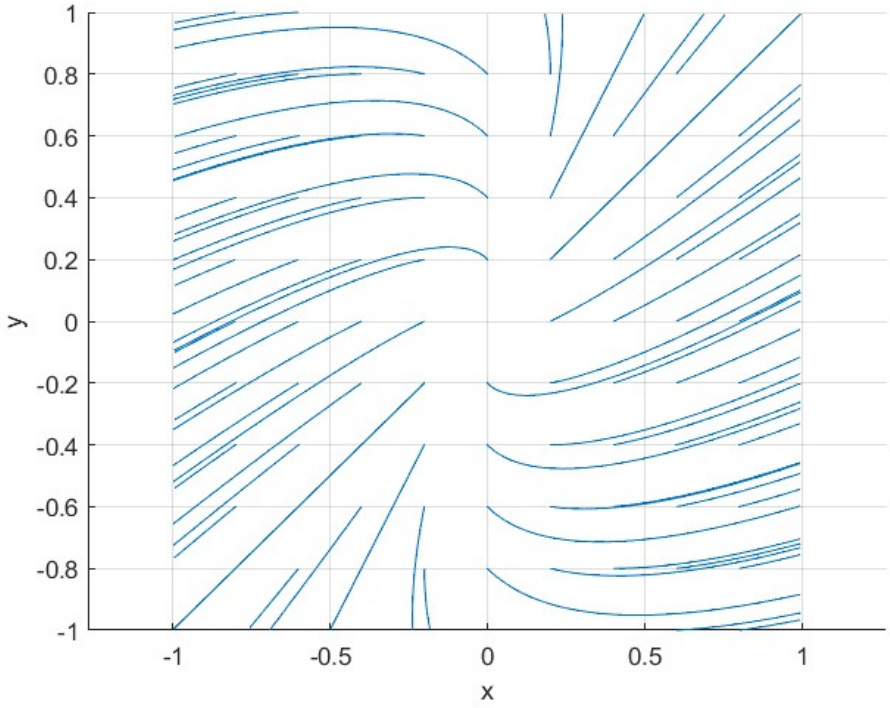


Figure 4: Phase portrait centered on the fixed point at the origin for the first linear system

as we can see, no matter how close the starting point is to (0,0), the system will diverge from it. We expected this behaviour because both eigenvalues are positive.

We can repeat the same procedure with the other systems.

To find an analytic solution to the initial value problem using the eigenvalue-eigenvector method for the system

$$x' = 3x - 2y,$$
$$y' = 2x + 3y$$

we need to find the eigenvalues and eigenvectors for the matrix

$$\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$$

The eigenvalues are $\lambda_1 = 3 + 2i$ and $\lambda_2 = 3 - 2i$ and the corresponding eigenvectors are $e_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$ and $e_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$

We can find the solution by studying

$$\mathbf{x}(t) = c_1 \begin{bmatrix} i \\ 1 \end{bmatrix} e^{(3+2i)t}$$

$$\mathbf{x}(t) = c \begin{bmatrix} i \\ 1 \end{bmatrix} e^{(3+2i)t} = c \begin{bmatrix} i \\ 1 \end{bmatrix} e^{3t} \Big(\cos(2t) + i \sin(2t) \Big) = c \begin{bmatrix} -\sin(2t) \\ \cos(2t) \end{bmatrix} e^{3t} + ic \begin{bmatrix} -\cos(2t) \\ -\sin(2t) \end{bmatrix} e^{3t}$$

This means that the solution is

$$\mathbf{x}(t) = c_1 \begin{bmatrix} -\sin(2t) \\ \cos(2t) \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} -\cos(2t) \\ -\sin(2t) \end{bmatrix} e^{3t}$$

We can also find a numerical solution to the initial value problem using a numerical ODE solver

```
function dydt = ode2(t, y)
   dydt = [3*y(1) - 2*y(2); 2*y(1) + 3*y(2)];
end

[t,y] = ode45(@ode2,[0 1],[a; b]);
```

where $[a; b]$ is the starting point.

We can compare the two solutions on a plot

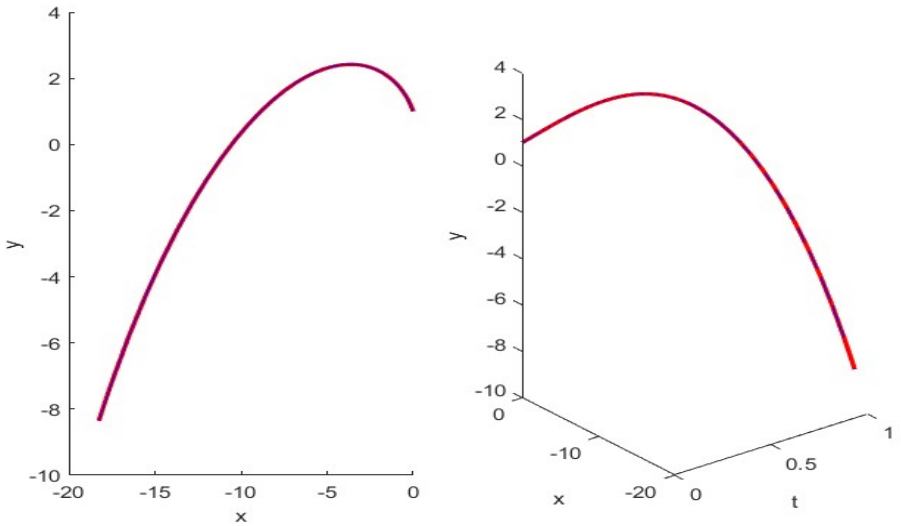


Figure 5: starting point (0,1)

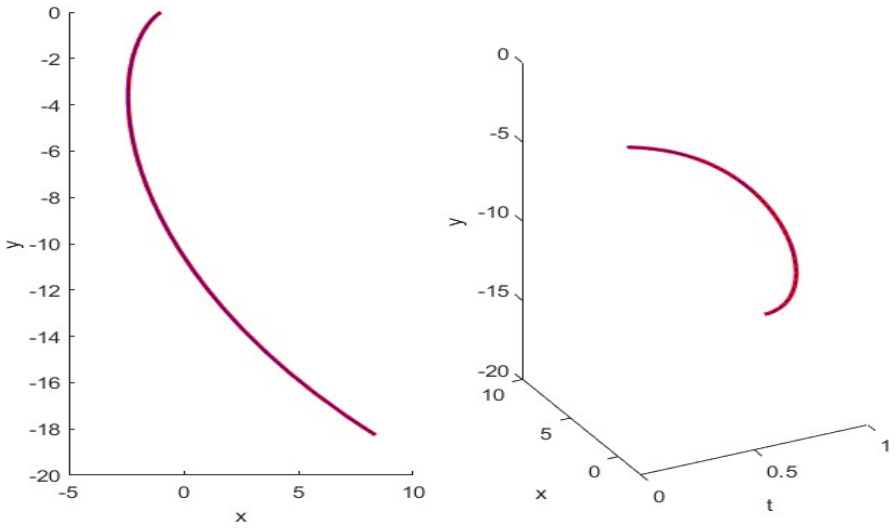


Figure 6: starting point (-1,0)

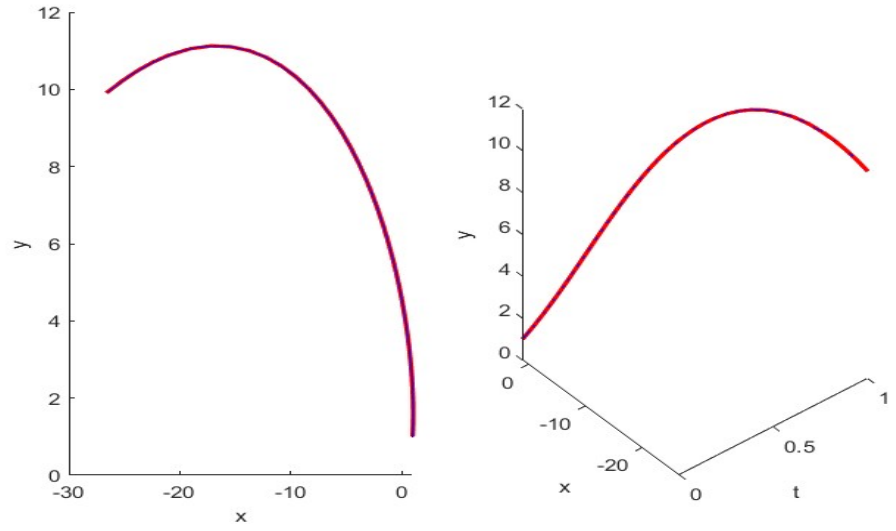


Figure 7: starting point (1,1)

As we can see, the solutions are exactly the same.

The phase portrait centered on the fixed point at the origin is in fig.8

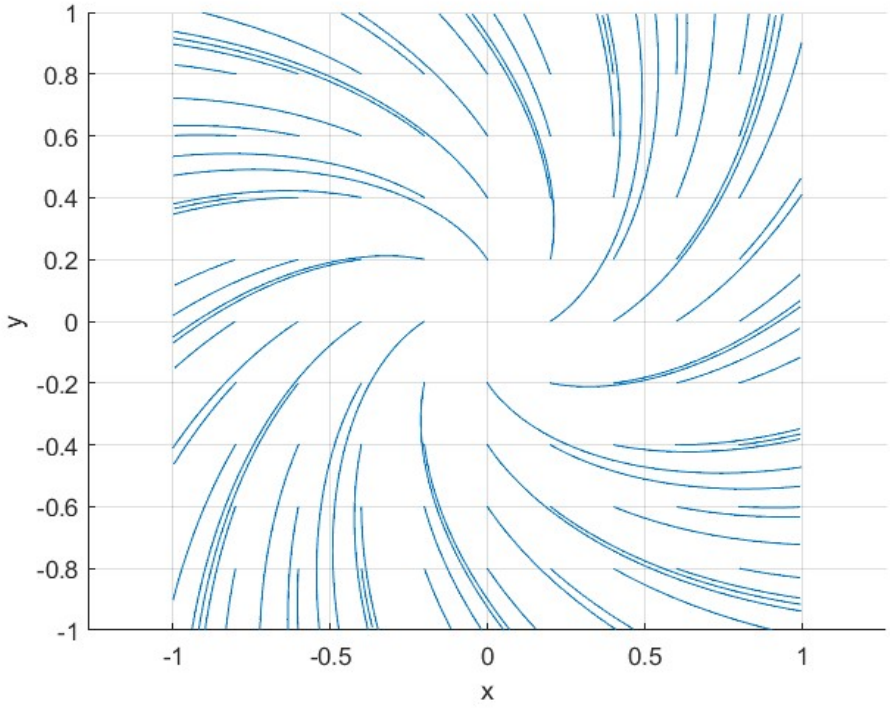


Figure 8: Phase portrait centered on the fixed point at the origin for the second linear system

as we can see, no matter how close the starting point is to (0,0), the system will diverge from it. We expected this behaviour because both eigenvalues have positive real part.

To find an analytic solution to the initial value problem using the eigenvalue-eigenvector method for the system

$$x' = x + y,$$
$$y' = -x + 3y$$

we need to find the eigenvalues and eigenvectors for the matrix

$$\begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix}$$

The eigenvalues are $\lambda_1 = \lambda_2 = 2$ and the corresponding eigenvector are $e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

because the eigenvalue has multiplicity 2, we also have to find $\begin{bmatrix} a \\ b \end{bmatrix}$ such that

$$\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

This means that the solution is

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) e^{2t}$$

We can also find a numerical solution to the initial value problem using a numerical ODE solver

```
function dydt = ode3(t, y)
    dydt = [y(1) + y(2); -1*y(1) + 3*y(2)];
end

[t,y] = ode45(@ode3,[0 1],[a; b]);
```

where $[a; b]$ is the starting point.

We can compare the two solutions on a plot

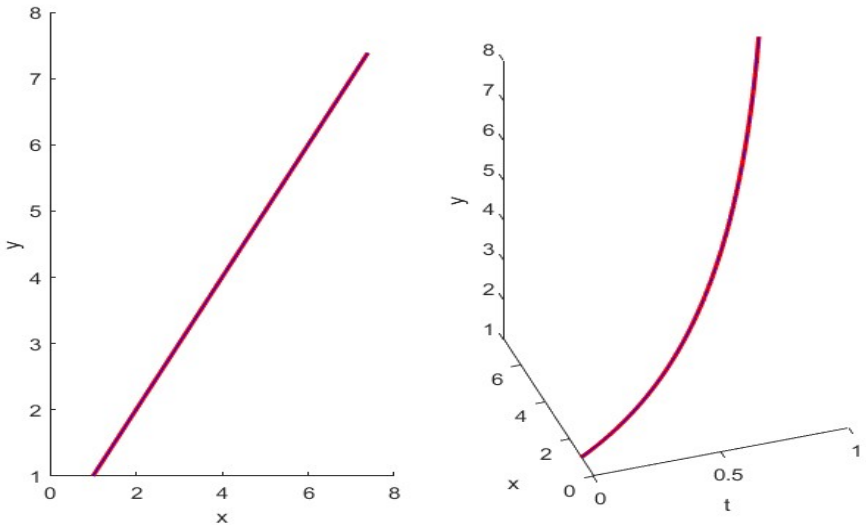


Figure 9: starting point (1,1)

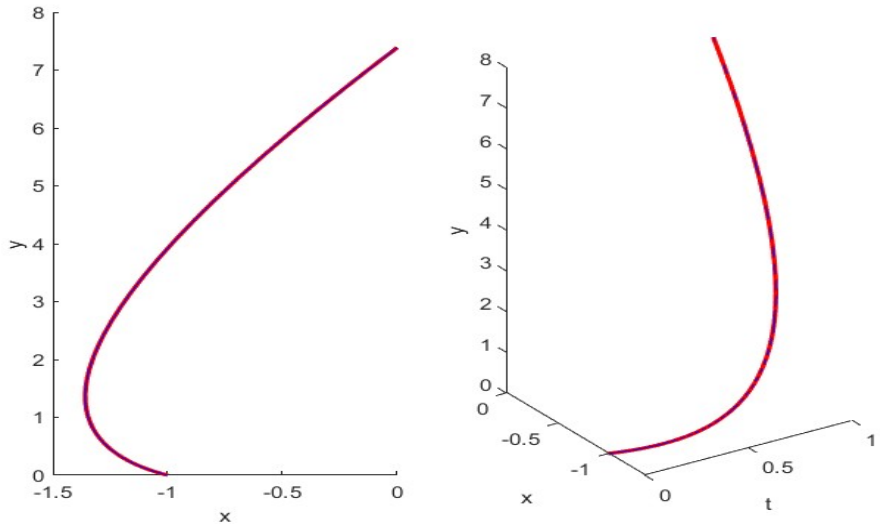


Figure 10: starting point (-1,0)

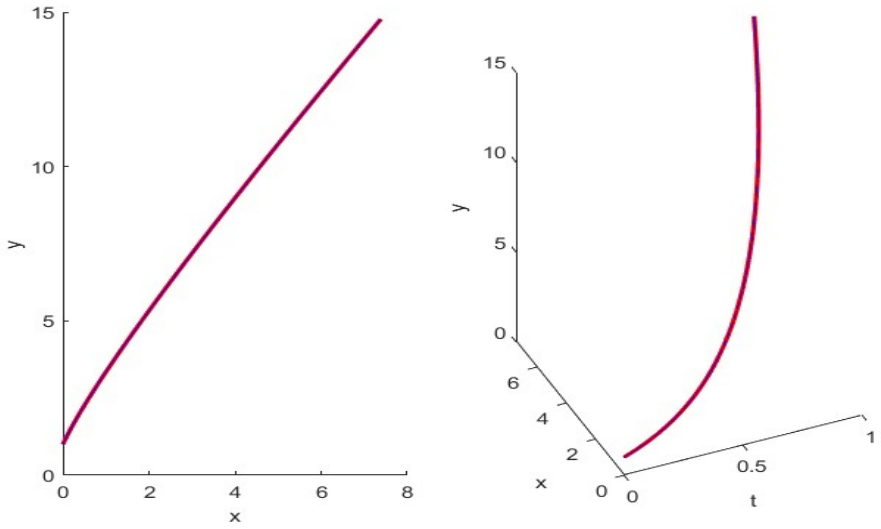


Figure 11: starting point (0,1)

As we can see, the solutions are exactly the same.

The phase portrait centered on the fixed point at the origin is in 12

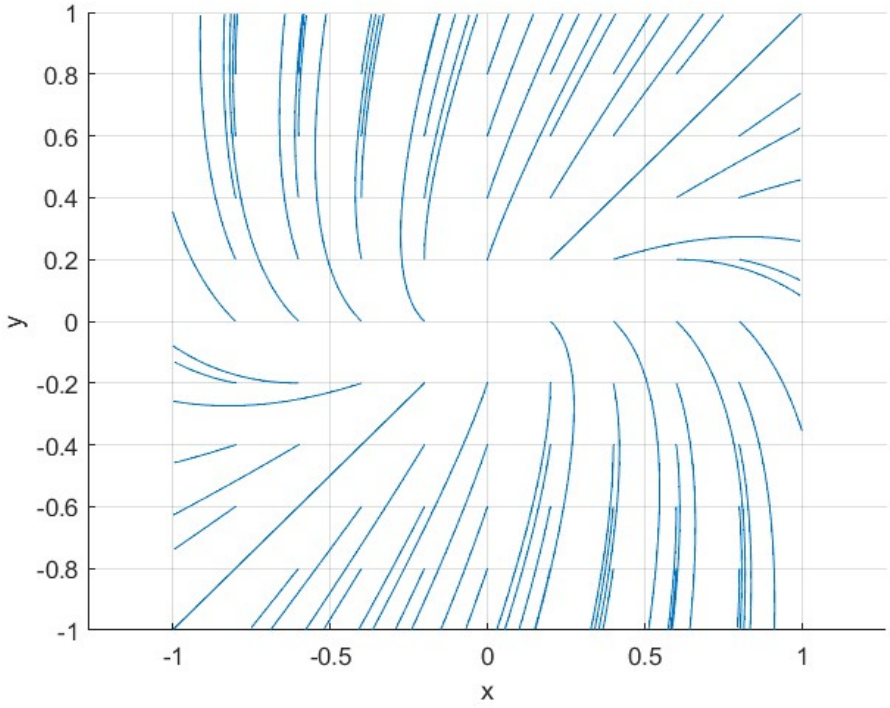


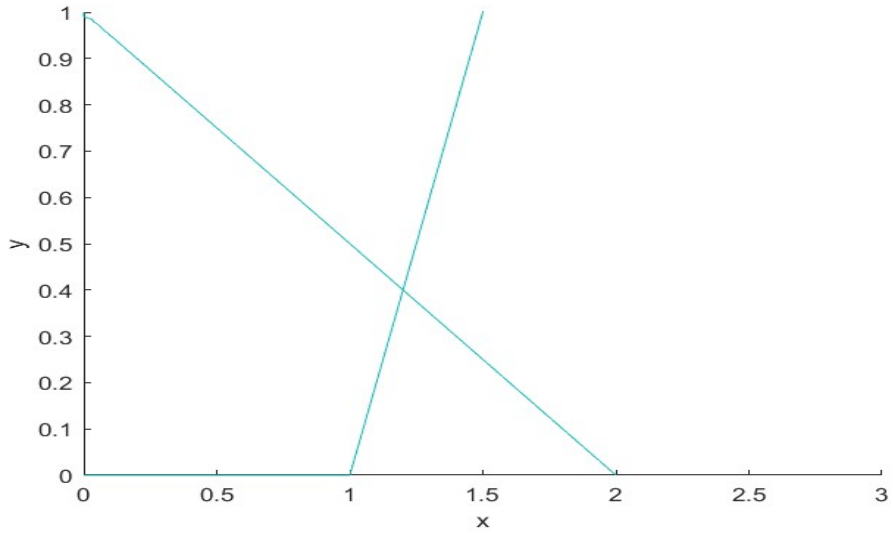
Figure 12: Phase portrait centered on the fixed point at the origin for the third linear system

as we can see no matter how close the starting point is to (0,0), the system diverge from it. We expected this behaviour because both eigenvalues are positive.

2)

We first consider $\alpha = \frac{1}{2}, \beta = 1$ and $\gamma = \frac{1}{2}$

Plotting the zero contours for the right hand side functions determining the Lotka-Volterra system we find



We get the point $(1.18, 0.404)$ as a rough estimation of a stable point, then using Newton's method

$$\mathbf{x}_{(t+1)} = \mathbf{x}_t - \mathbf{J}^{-1} f(\mathbf{x})$$

we find that the stable point is in $(1.20, 0.40)$

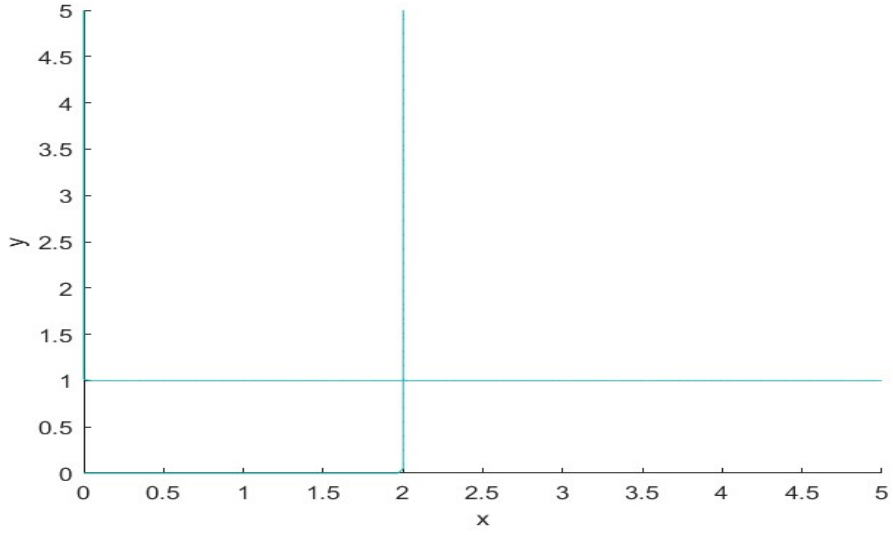
The linearized system around this stable point is

$$\mathbf{x}' = \begin{bmatrix} -0.6 & -1.2 \\ 0.4 & -0.2 \end{bmatrix} \mathbf{x}$$

We can investigate the stability of the fixed points by looking at the eigenvalues of $\begin{bmatrix} -0.6 & -1.2 \\ 0.4 & -0.2 \end{bmatrix}$ which are $\lambda_1 = -0.4000 + 0.6633i$ and $\lambda_2 = -0.4000 - 0.6633i$. The fixed point is asymptotically stable because both eigenvalues have negative real part.

Now considering $\alpha = 0, \beta = 2$ and $\gamma = 0$

Plotting the zero contours for the right hand side functions determining the Lotka-Volterra system we find



We get the point $(2.0, 1.06)$ as a rough estimation of a stable point, then using Newton's method

$$\mathbf{x}_{(t+1)} = \mathbf{x}_t - \mathbf{J}^{-1} f(\mathbf{x})$$

we find that the stable point is in $(2.00, 1.00)$
The linearized system around this stable point is

$$\mathbf{x}' = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \mathbf{x}$$

We can investigate the stability of the fixed points by looking at the eigenvalues of $\begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$ which are $\lambda_1 = i\sqrt{2}$ and $\lambda_2 = -i\sqrt{2}$. The fixed point is non-hyperbolic because both eigenvalues have zero as real part, the stability depends on the non-linear terms.

Part 2

1)

The derivation of the Stommel box model is based on the system shown in fig.13

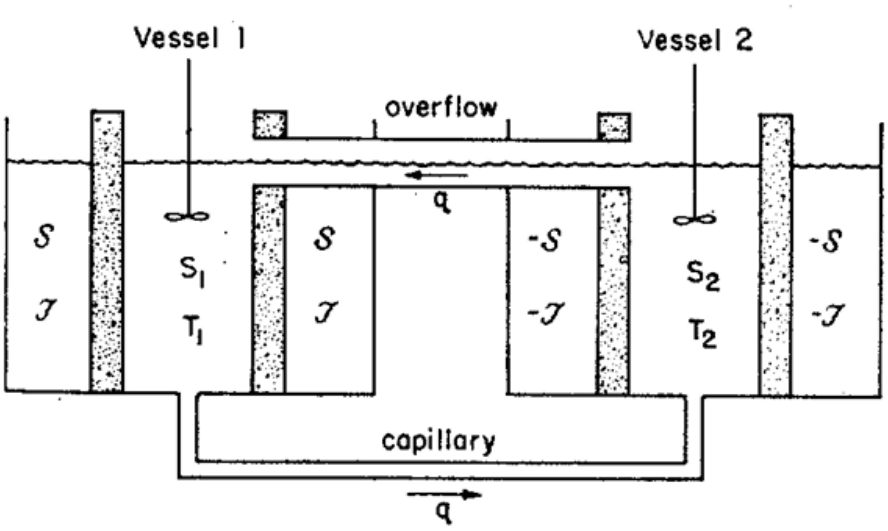


Figure 13: Stommel's two vessel experiment

It is formed by two stirred vessels separated by porous walls from two reservoirs. Because we are concerned with solutions, we can define $T = T_1 = -T_2$ and $S = S_1 = -S_2$. Also as we can see from the picture, Vessel 1 is surrounded by water at temperature $\mathcal{T}$ and salinity $\mathcal{S}$ while Vessel 2 is surrounded by water at temperature $-\mathcal{T}$ and salinity $-\mathcal{S}$.

The porous walls allow for the transfer of heat and salt in a linear fashion

$$\frac{dT}{dt} = c(\mathcal{T} - T)$$

$$\frac{dS}{dt} = d(\mathcal{S} - S)$$

Moreover, the two vessels are connected by a large overflow at the top so that the free surface of water is level across the top of each vessel and a capillary tube at the bottom whose resistance k is such that the flow q in the tube is directed from the high pressure (high density) vessel toward the low pressure (low density) vessel by a simple linear law

$$kq = \varrho_1 - \varrho_2$$

The flow q is positive going from vessel 1 to vessel 2, negative otherwise.
ϱ represents the equation of state

$$\varrho = \varrho_0 \left(1 - \alpha T + \beta S\right)$$

which means

$$kq = \varrho_0(1 - \alpha T_1 + \beta S_1 - (1 - \alpha T_2 + \beta S_2)) = 2\varrho_0(-\alpha T + \beta S)$$

Then the full model is

$$\frac{dT}{dt} = c(\mathcal{T} - T) - |q|(T - (-T)) = c(\mathcal{T} - T) - |2q|T$$

$$\frac{dS}{dt} = d(\mathcal{S} - S) - |q|(S - (-S)) = d(\mathcal{S} - S) - |2q|S$$

Note that the flow has an absolute value because it does not depend on the direction of the circulation.
We can scale the system using

$$\tau = ct, \quad \delta = \frac{d}{c}, \quad y = T/\mathcal{T}, \quad x = S/\mathcal{S}, \quad f = \frac{2q}{c}$$

$$\frac{dT}{dt} = c(\mathcal{T} - T) - |2q|T \Rightarrow c\mathcal{T}\frac{dy}{d\tau} = c\mathcal{T}(1-y) - c|f|\mathcal{T}y \Rightarrow \frac{dy}{d\tau} = 1 - y - |f|y$$

$$\frac{dS}{dt} = d(\mathcal{S} - S) - |2q|S \Rightarrow c\mathcal{S}\frac{dx}{d\tau} = c\delta\mathcal{S}(1-x) - c|f|\mathcal{S}x \Rightarrow \frac{dy}{d\tau} = \delta(1-x) - |f|x$$

and, defining $R = \frac{\beta\mathcal{S}}{\alpha\mathcal{T}}$ and $\lambda = \frac{ck}{4\varrho_0\alpha\mathcal{T}}$

$$f = \frac{2q}{c} = \frac{4\varrho_0}{ck}(-\alpha\mathcal{T}y + \alpha\mathcal{T}Rx) = \frac{4\varrho_0\alpha\mathcal{T}}{ck}(-y + Rx) = \frac{(-y + Rx)}{\lambda}$$

The final model is then

$$\begin{aligned}
\frac{dx}{dt} &= \delta(1-x) - |f|x, \\
\frac{dy}{dt} &= 1 - y - |f|y, \\
f &= \frac{1}{\lambda}(Rx - y)
\end{aligned}$$

2)

a)

Considering $\delta = \frac{1}{6}$, $\lambda = 0.2$ and $R = 2$ we can plot

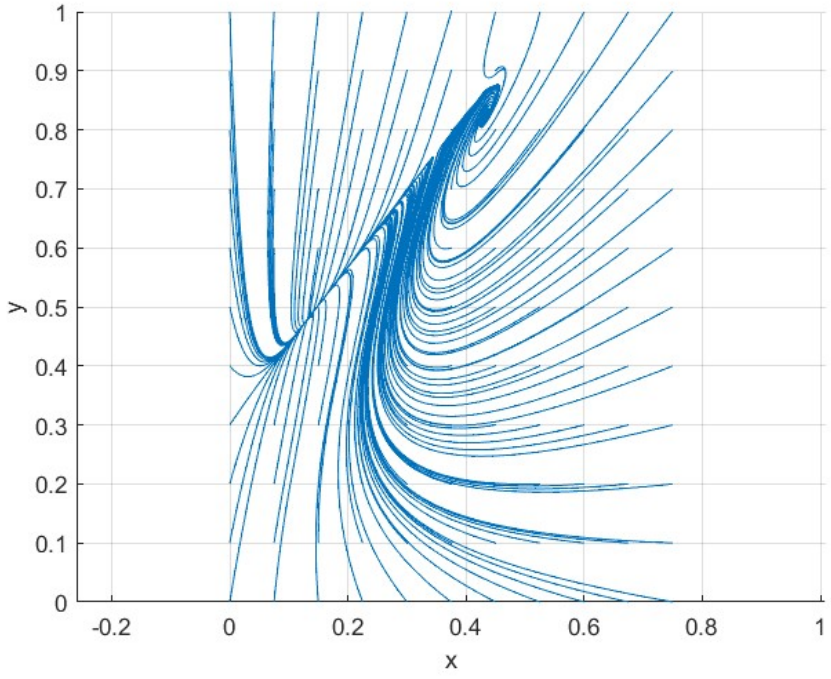


Figure 14: Rough visualization of the phase portrait of the system using a streamline plotting function for initial conditions in the domain [0, 0.75] × [0, 1].

We can see that from every starting point the system converges to one of two points. We can assume they are two asymptotically stable points, where, comparing them to the behaviour of fixed points in the Poincaré Diagram, one is a stable node and the other is a stable spiral. Between them we can see a saddle point. The position of the starting point relative to the stable variety of the saddle node decides in which stable point the system will converge.

b)

We can find the fixed points by first getting a rough location with a contour plot

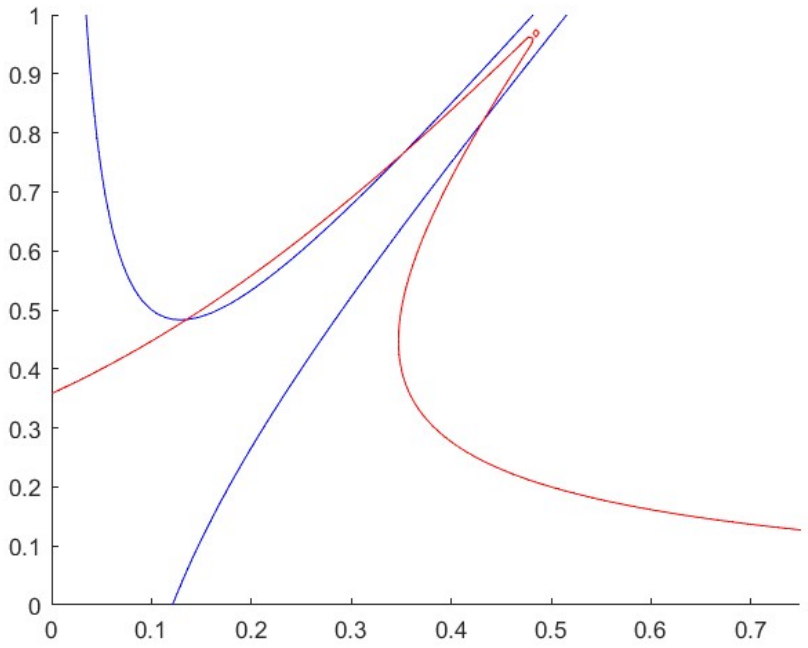


Figure 15: Zero contours of the right hand side of the two equations defining the Stommel system

We find three equilibrium points that are approximately $(0.129, 0.485)$, $(0.356, 0.778)$, $(0.432, 0.828)$ and then use Newton iteration to find high precision numerical values for the locations of the fixed points, that are $\bar{\mathbf{x}}_1 = (0.1350, 0.4836)$, $\bar{\mathbf{x}}_2 = (0.3518, 0.7651)$, $\bar{\mathbf{x}}_3 = (0.4321, 0.8203)$

c)

The matrix of the linearized system for $\bar{\mathbf{x}}_1$ is

$$\begin{bmatrix} 0.1154 & -0.6750 \\ 4.8358 & -4.4858 \end{bmatrix}$$

and the eigenvalues are $\lambda_1 = -0.7609, \lambda_2 = -3.6095$. They are both negative real values $\Rightarrow \bar{\mathbf{x}}_1$ is a stable node.

The matrix of the linearized system for $\bar{\mathbf{x}}_2$ is

$$\begin{bmatrix} 3.0448 & -1.7592 \\ 7.6510 & -5.1325 \end{bmatrix}$$

and the eigenvalues are $\lambda_1 = 0.7609, \lambda_2 = -2.8486$. They are both real, but one of them is positive while the other one is negative $\Rightarrow \bar{\mathbf{x}}_2$ is a saddle.

The matrix of the linearized system for $\bar{\mathbf{x}}_3$ is

$$\begin{bmatrix} -4.7063 & 2.1603 \\ -8.2028 & 2.8823 \end{bmatrix}$$

and the eigenvalues are $\lambda_1 = -0.9120 + 1.8231i, \lambda_2 = -0.9120 - 1.8231i$. They are both imaginary numbers with negative real part $\Rightarrow \bar{\mathbf{x}}_3$ is a stable spiral.

d)

As we saw in the previous point, the second equilibrium is a saddle, which means that it has both a stable and an unstable manifold. For λ_1 the eigenvector is $e_1 = \begin{bmatrix} 0.6102 \\ 0.7922 \end{bmatrix}$, for λ_2 the eigenvector is $e_2 = \begin{bmatrix} 0.2860 \\ 0.9582 \end{bmatrix}$.

We can look at the behaviour of the system starting from $\bar{x}_2 + \varepsilon \mathbf{e}_1$, $\bar{x}_2 - \varepsilon \mathbf{e}_1$ in the future(because this is the unstable manifold) and from $\bar{x}_2 + \varepsilon \mathbf{e}_2$, $\bar{x}_2 + \varepsilon \mathbf{e}_2$ in the past(because this is the stable manifold), where $\epsilon = 10^{-9}$.

The resulting plots are

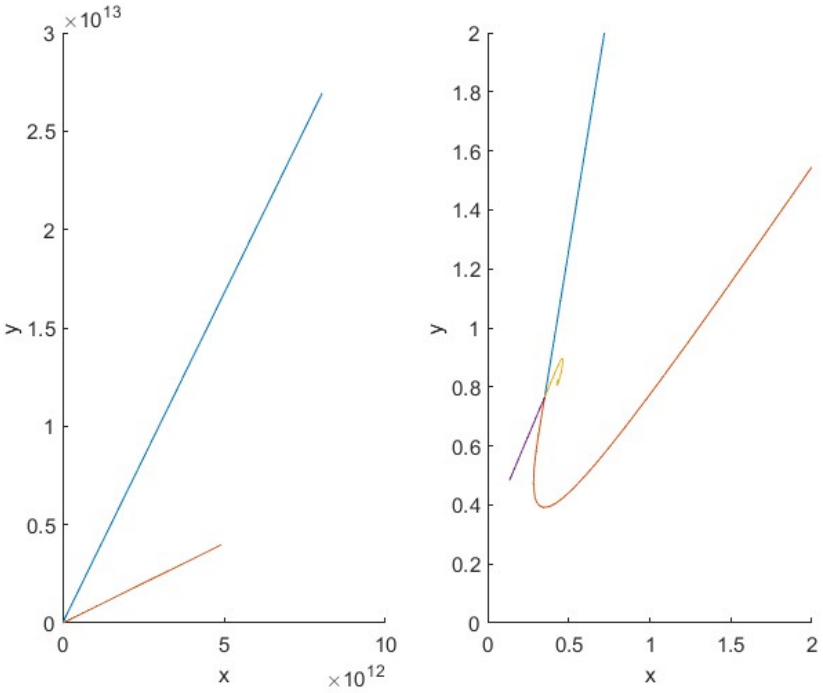


Figure 16: Portions of the stable and unstable manifolds of the saddle point, on the right the zoomed plot to see the behaviour of the unstable manifold converging in one of the stable points

e)

Given

$$g(\lambda, x, y) = \delta(1 - x) - \left|\frac{Rx - y}{\lambda}\right| x$$

$$h(\lambda, x, y) = 1 - y - \left|\frac{Rx - y}{\lambda}\right| y$$

we define

$$u = (g_\lambda, g_x, g_y)$$

$$v = (h_\lambda, h_x, h_y)$$

and we can build the bifurcation diagram solving the ODE

$$(\lambda', x', y') = \frac{u \times v}{\|u \times v\|}$$

using as starting points the three stationary points found earlier. The resulting plot is fig.17

f)

From the calculations used earlier we have that $\lambda_c = 0.3338$, and the hyperbolic stationary point is $\mathbf{x} = (0.2561, 0.6738)$.

Here the linearized matrix is $\begin{bmatrix} 0.8838 & -0.7673 \\ 4.0372 & -3.5027 \end{bmatrix}$. One of its eigenvalues is indeed 0.

g)

We can now use linearization to investigate stability of the branches found in point e). The result of this calculation is in fig.18

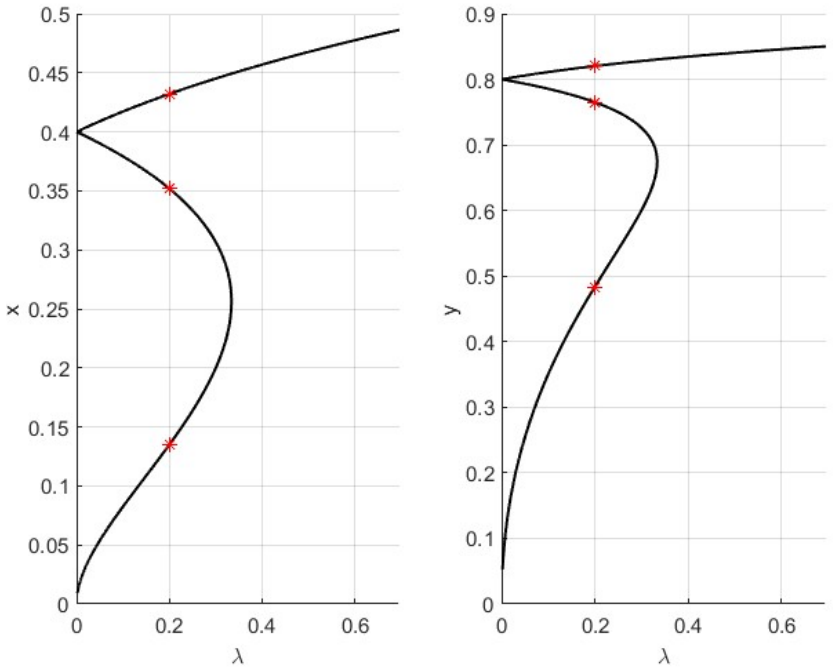


Figure 17: Bifurcation plot of the Stommel system, on the left on the (λ, x) plane, on the left on the (λ, y) plane. The red stars represent the starting points used for the ODE

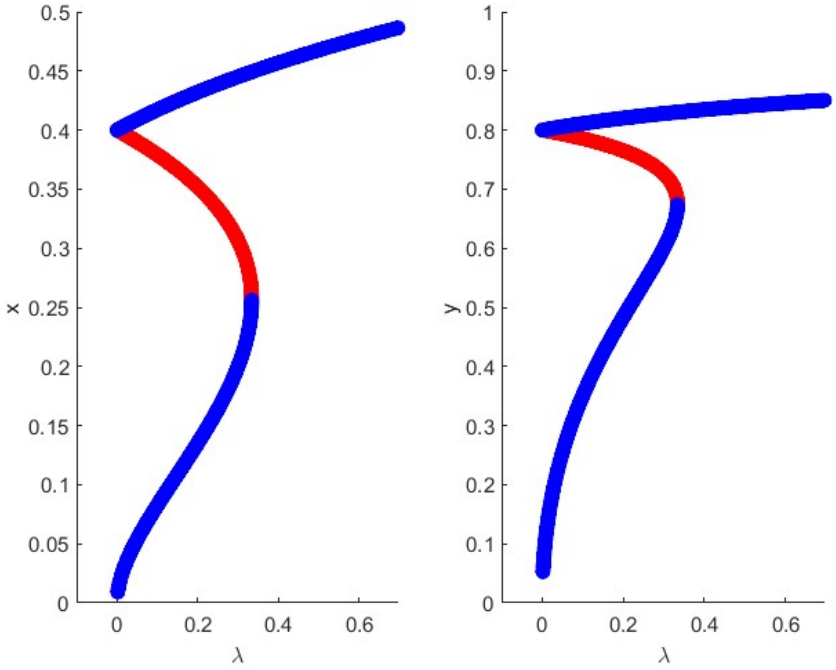


Figure 18: Bifurcation plot of the Stommel system, red represents unstable fixed points, blue represents stable fixed points